SILVERARROWS FLIGHT ACADEMY • SYSTEMS ARCHITECTURE & GAP ANALYSIS REPORT

SilverArrows vs. FlightLogger REA Specification

Exhaustive 3-Way Difference Audit: Current Application Codebase vs. 13 FlightLogger REA .md Blueprints

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Scope: (1) Exclusive to SilverArrows | (2) Missing from FlightLogger REA Spec | (3) Missing in Both Systems

Executive Quantitative Comparison Matrix

This report presents a line-by-line architectural comparison between the live SilverArrows Aviation Academy codebase (/src, /supabase) and the 13 reverse-engineered FlightLogger REA specification files located in /flightloggerreatask (00_MASTER_REA_INDEX.md through 11_UI_AND_BACKEND_COMPLETE_RECREATION_CHECKLIST.md).

Architectural Dimension	Current SilverArrows Application	FlightLogger REA .md Specification	Gap Severity
Database Entities	19 synced tables (15 operational + 4 approval, profile, learning & audit tables)	32 core relational entities (+ SMS hazard reports & ACI invoicing tables)	**HIGH (19+ missing tables)**
Application Routes	17 routes (1 landing, 1 E-Learning portal, 1 auth, 14 _shell workspace routes)	60+ routes (Global /my SSO hub + path-scoped /a/:tenant/... multi-tenant routes)	**HIGH (43+ missing views)**
API & RPC Operations	25 store actions + Supabase REST/Realtime diff sync (mapping.ts / sync.ts)	144 cataloged GraphQL Queries, Mutations & ActionCable Subscriptions	**HIGH (ACID RPCs needed)**
RBAC Role Matrix	4 roles (admin, instructor, ground, student) — scalar 1 role per user	7 roles (admin, staff, instructor, crew, student, renter, guest) — multi-role per user	**MEDIUM (3 missing roles)**
Flight Lifecycle FSM	3 states (scheduled, completed, cancelled) + Admin Post-Flight Yes/No prompt	6-stage FSM (scheduled -> briefing -> flight -> landed -> debriefing -> completed)	**CRITICAL (Core Ops FSM)**
Post-Flight Commit	Sequential client store updates (flights, aircraft.hours, completions, bookings)	6-Way Atomic ACID Stored Procedure (fn_register_flight) with full rollback	**CRITICAL (Data Integrity)**
Pre-Flight Dispatch Gating	3 checks (time overlap, active member status, aircraft grounded/overdue status)	7-point regulatory gate (Medical, 90d currency, Class rating, Maint hours vs duration, FTL 12h rest & 7/28/365d rolling limits, 24h Solo sign-off)	**CRITICAL (Safety Gating)**
Syllabus & Grading Model	Lesson-level Pass/Fail (completed/incomplete) + 1-100 score + text notes	EASA/ICAO CBTA 1-5 Norm Matrix across maneuvers (master_exercises) & 8 Core Competencies (KNO, FPA, FPO, COM, WLM, SAW, PRB, LST)	**HIGH (CBTA Compliance)**
Ground School & Textbooks	Full interactive digital textbook reader, hotspot diagrams, notes, quizzes & 3D/SVG Atlas	Only classroom schedule slots (lectures) & theory exam score entry (exam_results)	**SILVERARROWS ADVANTAGE**
Regulatory Authority Target	Indian DGCA (Computer No, eGCA ID, SPL/FRTOL/RTR, 5 DGCA exams) + EASA FTL	European EASA Part-FCL / Part-ORA / Part-FTL / Part-ML & FAA Part 141	**COMPLEMENTARY**

Source Files Audited in /flightloggerreatask

File Name	Specification Focus & Architectural Contents
00_MASTER_REA_INDEX.md	Master topology index linking bundle AST analysis, 144 operations, 32 database tables, and 7 core domains.
01_BUNDLE_AST_2_5M_ANALYSIS.md	2.54M-char production Webpack/Babel AST decompilation (React 18, Apollo Client, Redux-Persist, FullCalendar, D3).
02_GRAPHQL_CATALOG_144_OPERATIONS.md	Complete registry of 144 GraphQL queries, mutations, and subscriptions across 9 operational domains.
03_DATABASE_SCHEMA_32_ENTITIES.md	DDL blueprint for 32 PostgreSQL tables across 7 functional domains with foreign keys and check constraints.

File Name	Specification Focus & Architectural Contents
04_TENANT_ROUTES_AND_PAGE_DIRECTORIES.md	60+ routes spanning global /my SSO pilot portal and tenant-scoped /a/:tenant/... academy operations.
05_CORE_ENGINES_AND_STATE_MACHINES.md	6-Stage Flight FSM, 6-Way Atomic Post-Flight Commit (fn_register_flight), CBTA 1-5 Grading, and 7-Point Dispatch Gate.
06_REALTIME_WEBSOCKETS_AND_STATE_PERSISTENCE.md	ActionCable / Supabase Realtime channels and crash-proof SessionStorage buffer (persist:userCustomLecture).
07_ROLES_PERMISSIONS_AND_PORTALS.md	7-Tier RBAC matrix (Admin, Staff, Instructor, Crew, Student, Renter, Guest) and RLS security policies.
08_AGENT_SKILLS_AND_INTEGRATION_LINKS.md	Stripe ACI billing, ADS-B telemetry auto-landing, S3 presigned document vault, SendGrid/Twilio, and WebAuthn 2FA.
09_COMPREHENSIVE_TEST_SPECIFICATION_AND_MATRIX.md	100+ unit, integration, E2E, and chaos fault-injection test vectors for aviation regulatory invariants.
10_DATABASE_TABLES_COLUMNS_AND_USAGES_ENCYCLOPEDIA.md	Column-by-column encyclopedia detailing exact SQL types, triggers, and UI bindings for all 32 tables.
11_UI_AND_BACKEND_COMPLETE_RECREATION_CHECKLIST.md	Screen-by-screen UI recreation specification (Gantt scheduler, Wallboard, 4-tab Debrief Wizard, Radar Charts).

SECTION 1 OF 3 • SILVERARROWS EXCLUSIVE CAPABILITIES

1. What Is in Your App Today, But MISSING in the FlightLogger .md Files

Custom-built E-Learning textbooks, Indian DGCA compliance, multi-stage admin approval workflows, and live weather synthesis

Your current SilverArrows codebase contains 11 rich functional subsystems that are completely absent from the FlightLogger REA .md specification files. These must be preserved intact when expanding the platform.

#	SilverArrows Exclusive Feature	Source Files in Your App	Detailed Capability Absent from FlightLogger REA .md Spec
1.1	**Interactive E-Learning Digital Textbook & Hotspot Reader**	/elearning, /learning/$chapterId, airframes-content.ts, electrics-content.ts, powerplant-content.ts, airlaw-content.ts, learning-illustrations.ts	FlightLogger REA only tracks classroom schedule slots (lectures) and exam scores. SilverArrows provides a full digital ground-school textbook with verbatim & expanded markdown chapters, generated technical illustrations with interactive static coordinate hotspots, personal study notes, bookmarks, and inline chapter quizzes persisted in learning_records.
1.2	**Admin Ground-School Course Assignment Console**	src/components/assigned-courses-panel.tsx, assignedCourseIds on tenant_members	Allows the Administrator to selectively assign or revoke specific ground-school textbook courses per student, emitting real-time course_assigned and course_unassigned notifications.
1.3	**Interactive 3D / SVG Aviation Technical Atlas**	src/routes/_shell.atlas.tsx, src/components/atlas/, src/lib/atlas/	Dedicated interactive visual aircraft systems exploration atlas inside the authenticated workspace shell.
1.4	**Cinematic Takeoff Sequence, Terrain Compass & Live IST Clock**	src/components/homepage-aviation-sequence.tsx, live-ist-clock.tsx, network-time.ts	Interactive landing page featuring an animated aircraft takeoff sequence, real-time mountain/airstrip terrain compass map (VIDP 28R/10L), 3D tactile gateway cards, and a network-synced Indian Standard Time (IST) cockpit clock.
1.5	**Indian DGCA Regulatory Pilot Credentials & Written Exams**	src/lib/types.ts, src/lib/api/mapping.ts, src/routes/_shell.crew.tsx	Native support for Indian DGCA fields: Computer Number, eGCA ID, File Number, SPL (Student Pilot License), FRTOL, RTR(A) numbers/expiries, and 5 DGCA written exam statuses (Air Regulation, Air Navigation, Meteorology, Technical General, Technical Specific).
1.6	**Strict Registration Gate & Real-Time Admin Login Lock**	src/routes/login.tsx, isd-codes.ts, PENDING_ADMIN_APPROVAL	Enforces @flysilverarrows.com usernames, ISD country code phone validation (default +91), atomic uniqueness across username/email/phone, sole-admin lock (admin@flysilverarrows.com), and a real-time Account Pending Administrator Login Approval screen.
1.7	**Student Flight Booking Approval Queue & Audit Trail**	src/routes/_shell.admin.approvals.tsx, approval-audit-trail.tsx, DB triggers	Database triggers force student-created flight bookings into approval_status = 'pending'. The Admin Approvals Console provides one-click Approve/Decline and an immutable chronological Approval Audit Trail.
1.8	**Automated Post-Flight 'Did the Flight Happen?' Admin Prompt**	src/components/post-flight-prompt.tsx, confirmPostFlightOutcome()	Detects past scheduled bookings whose end time has elapsed and prompts the Administrator to verify whether the flight occurred (Yes auto-creates the flight log & updates aircraft hours; No cancels the slot and records a DNF audit entry).
1.9	**Student Flight Log Verification & Profile Change Diff Queue**	submitStudentFlightLog(), verifyFlightLog(), profile_requests, profile-requests.tsx	Students can submit flight logs in 'pending' status for Instructor/Admin verification. Non-admin profile edits create allowlisted before/after diffs (changes vs before_values) requiring Admin approval before mutating tenant_members.
1.10	**Live Open-Meteo Aerodrome Weather & METAR/TAF Synthesizer**	src/lib/weather.ts, src/components/weather-widget.tsx	Fetches real-time aerodrome temperature, wind vector/gusts, visibility, cloud cover, and QNH pressure, synthesizes aviation METAR/TAF strings and flight categories (VFR/MVFR/IFR/LIFR), and attaches immutable weather snapshots to bookings and flight logs.
1.11	**Pilot Proficiency Badges, Role Switcher & Privacy Telemetry**	proficiency-badges.tsx, view-as.tsx, _shell.database.tsx, privacy-tracker.ts	Computes Bronze/Silver/Gold tiers across 8 pilot badges and Pilot Ranks (Cadet Aviator to Master Command Aviator), plus an in-app Role Impersonation Switcher, Live Database Table Inspector, and session activity/idle telemetry tracker.

SECTION 2 OF 3 • CRITICAL GAPS FROM FLIGHTLOGGER REA BLUEPRINT

2. What Is MISSING in Your App from the FlightLogger .md Files

Detailed breakdown of 20 missing database tables, 3 core backend state engines, 43+ routes, CBTA 1-5 grading, ACI billing & SMS

2.A — Missing Database Tables (19 Current Synced vs. 32 Core Entities in 03 & 10 .md)

Your current app syncs 19 tables in src/lib/api/mapping.ts (tenant_members, member_roles, aircraft, maintenance_items, programs, program_lessons, theory_subjects, classrooms, bookings, flights, completions, change_requests, exam_results, lectures, duty_records, profile_requests, learning_records, notification_reads, activity_feed). Below are the 20 relational tables defined in 03_DATABASE_SCHEMA_32_ENTITIES.md and 10_DATABASE_TABLES_COLUMNS_AND_USAGES_ENCYCLOPEDIA.md that are missing from your database:

Spec #	Missing Table Name	Domain	Key Columns, Constraints & Operational Purpose in FlightLogger REA
Table 5	**webauthn_credentials**	Auth & 2FA	credential_id, public_key, sign_count, device_name. Stores FIDO2 / Passkey hardware keys for biometric 2FA and logbook signing.
Table 6	**aircraft_models**	Fleet & CAMO	make, model_name, class_rating (SEP_LAND, MEP_LAND, TMG, SET, MET), engine_type, usable_fuel_liters, cruise_speed_knots, has_retractable_gear, has_variable_pitch_prop.
Table 9	**aircraft_remarks**	Fleet & CAMO	reported_by, flight_id, severity (info, warning, grounding_defect), status (open, deferred_mel, rectified), mel_reference, rectified_by_license (CAMO CRS sign-off).
Table 10	**certificate_types**	Compliance	name, category (medical, license, rating, radio, language_proficiency, passport, company_check), warning_days_before, blocks_booking, blocks_solo.
Table 11	**user_certificates**	Compliance	user_id, certificate_type_id, certificate_number, issue_date, expiry_date, file_storage_path (S3/Supabase Storage), verified_by. Replaces hardcoded flat columns.
Table 12	**certificate_warnings**	Compliance	user_id, user_certificate_id, warning_code (e.g., 'ANS: Medical (1) Missing', '90_DAY_LANDING_CURRENCY_LAPSED'), severity (expiring_soon, expired, missing_mandatory).
Table 14	**program_revisions**	Syllabus & CBTA	program_id, revision_number, effective_date, approved_by_authority_ref, is_current. Allows syllabus version upgrades without breaking active student cohorts.
Table 15	**program_phases**	Syllabus & CBTA	revision_id, name (e.g. Phase 1: Pre-Solo), sequence_order, requires_stage_check. Enforces stage-check barriers between training phases.
Table 17	**master_exercises**	Syllabus & CBTA	lesson_id, code (e.g. '4.1.a'), title (e.g. 'Steep Turns 45 deg Bank'), norm_target_grade (1-5), is_mandatory. Granular maneuvers inside each syllabus lesson.
Table 18	**core_competencies**	Syllabus & CBTA	code (KNO, FPA, FPO, COM, WLM, SAW, PRB, LST), name, description, min_acceptable_score (default 2). The 8 ICAO Doc 9995 / EASA CBTA dimensions.
Table 19	**competency_indicators**	Syllabus & CBTA	competency_id, indicator_code (e.g. 'FPA.1'), description. Observable Behavioral Indicators (OBIs) checked by instructors during post-flight debriefing.
Table 20	**user_programs**	Syllabus & CBTA	student_id, program_id, revision_id, primary_instructor_id, enrolled_at, completed_at, status (active, on_hold, completed, withdrawn).
Table 21	**user_lectures**	Syllabus & CBTA	user_program_id, lesson_id, flight_id, instructor_id, attempt_number, status (planned, in_progress, passed, repeat_required), instructor_signature_hash, student_signature_hash.
Table 22	**user_exercises**	Syllabus & CBTA	user_lecture_id, exercise_id, grade (CHECK 1..5), attempt_count, instructor_remark. Stores the 1-5 norm grade for every maneuver flown.
Table 23	**graded_competencies**	Syllabus & CBTA	user_lecture_id, competency_id, score (CHECK 1..5), observed_indicators[], debrief_comment. Feeds the 8-axis student CBTA radar chart.
Table 25	**booking_conflicts**	Dispatch & Ops	booking_id, conflict_type (aircraft_overlap, instructor_overlap, maintenance_overdue, certificate_expired, ftl_violation, weather_minima), is_blocking, overridden_by, override_reason.
Table 27	**landings**	Flight Ops	flight_id, airport_icao, day_full_stop, day_touch_and_go, night_full_stop, night_touch_and_go, approaches_ils, approaches_rnp, approaches_visual. Powers 90-day currency.
Table 28	**electronic_logbook_entries**	Flight Ops	user_id, flight_id, role_capacity (PIC, DUAL_STUDENT, INSTRUCTOR, SPIC, PICUS, SIM_INSTRUCTOR), total_flight_time, signature_hash, is_locked. Dual rows per flight.
Tables 30-32	**customers, price_lists, price_list_items, activities, invoices, invoice_items**	Finance (ACI)	Automated Customer Invoicing (ACI) ledger: customer account_balance & credit_limit, wet/dry aircraft hourly rates, unbilled flight activities, and finalized PDF/Stripe invoices.
SMS Ext.	**sms_reports / hazard_logs**	Safety (SMS)	ICAO Doc 9859 occurrence reporting: category, severity (1-5), likelihood (A-E), risk_index (e.g. '4B'), is_confidential, capa_mitigation_plan, status.

2.A (Cont.) — Missing Columns on Existing SilverArrows Tables

Existing Table	Current Columns in SilverArrows	Missing Columns Required by FlightLogger REA Specification
aircraft	id, registration, type, status, hours, next_due_hours, last_100hr, annual_due	model_id (FK to aircraft_models), serial_number, home_base_icao, meter_type (HOBBS_DECIMAL / TACH_DECIMAL / FLIGHT_TIME_MINUTES), current_tach_hours, current_hobbs_hours, total_airframe_hours, total_landings, status = 'maintenance_due'.
maintenance_items	id, aircraft_id, title, due_hours, due_date, status	interval_days, interval_landings, warning_hours_threshold, warning_days_threshold, next_due_landings, is_critical_grounding (auto-grounds tail at <= 0h), completed_by_camo_ref, work_order_ref.
bookings	id, aircraft_id, student_id, instructor_id, start_time, end_time, lesson_code, status, approval_status	booking_type (dual_training, solo_training, stage_check, skill_test, rental, charter, maintenance, ground_school), 6-stage status (briefing, flight, landed, debriefing), departure_icao, arrival_icao, alternate_icao, briefing_start_time, debriefing_end_time.
flights	id, date, aircraft_id, student_id, instructor_id, duration_hours, flight_type, lesson_code, day_hours, night_hours, instrument_hours, landings	4-point timestamps (off_block_time, takeoff_time, landing_time, on_block_time), tach_start, tach_stop, hobbs_start, hobbs_stop, fuel_start_liters, fuel_added_liters, fuel_stop_liters, oil_added_quarts, time_spic, time_picus, time_vfr, time_ifr, time_sim_fnpt, instructor_signed_at, student_signed_at.
duty_records	id, instructor_id, date, duty_hours, flight_hours, description	duty_start (TIMESTAMPTZ), duty_end (TIMESTAMPTZ), duty_type (flight_duty, ground_instruction, simulator_instruction, office_admin, standby_reserve), split_duty_break_minutes, fdp_hours, max_allowed_fdp_hours, mandatory_rest_until (TIMESTAMPTZ).

2.B — Missing Core Backend Engines & State Machines (05_CORE_ENGINES_AND_STATE_MACHINES.md)

- Engine 1: The 6-Stage Flight Lifecycle Finite State Machine (FSM): Your app only tracks 3 booking states (scheduled, completed, cancelled). The REA specification requires a 6-stage operational FSM: scheduled -> briefing -> flight -> landed -> debriefing -> completed. Transitioning scheduled -> briefing checks that no crew member has blocking certificate warnings and stamps briefing_start_time. Transitioning briefing -> flight verifies aircraft.status == 'airworthy', validates all 7 pre-flight dispatch gates, and stamps off_block_time & takeoff_time. Transitioning flight -> landed enforces landing_time > takeoff_time and blocks cancelling an airborne aircraft (IllegalStateTransitionError). Transitioning debriefing -> completed executes fn_register_flight.

- Engine 2: The 6-Way Atomic Post-Flight Commit (fn_register_flight ACID Stored Procedure): In your app, logging a flight updates client memory and syncs 4 tables sequentially. In the REA specification, completing a flight runs a single PostgreSQL stored procedure (fn_register_flight) that atomically commits 6 subsystems with full rollback on error: (1) Updates aircraft tach/hobbs/airframe hours, total_landings, fuel burn & oil uplift, decrements maintenance_items.hours_left, and auto-grounds the aircraft (status = 'maintenance_due') if any critical task hits <= 0h; (2) Inserts CBTA 1-5 grades into user_lectures, user_exercises, and graded_competencies, marking repeat_required and locking the next syllabus lesson if any mandatory grade is <= 2; (3) Synthesizes TWO immutable rows in electronic_logbook_entries (Student DUAL/PIC row + Instructor INSTRUCTOR/PIC row) sealed with SHA-256 signature_hash; (4) Closes the instructor's duty_records row and stamps mandatory_rest_until = NOW() + GREATEST(12h, duty_duration); (5) Calculates Automated Customer Invoicing (ACI) charges into activities (unbilled) and deducts from customers.account_balance; (6) Auto-spawns a draft SMS hazard report if an incident was flagged.

- Engine 3: The 7-Point Pre-Flight Dispatch Gating Engine: Your current bookingProblem() in src/lib/logic.ts checks only 3 conditions (time overlap, active user status, and whether aircraft maintenance is already <= 0). It is missing 5 critical regulatory gates: Gate 1 (Crew Medical, License, Radio & Language Proficiency expiry vs booking.end_time); Gate 2 (EASA FCL.060 90-day 3-landing passenger/solo currency query on landings); Gate 3 (Aircraft Model Class/Type rating endorsement check for SEP_LAND, MEP_LAND, retractable gear, variable-pitch prop); Gate 4 (Planned flight duration vs remaining maintenance hours: hours_left >= planned_flight_duration so a 2.5h flight cannot launch with 1.8h left before a 100h check); Gate 5 & 6 (EASA Part-FTL 12h mandatory_rest_until lock + rolling cumulative ceilings: max 60h duty in 7 days, 190h duty / 100h flight in 28 days, 900h flight in calendar year, 1000h flight in 365 days); Gate 7 (24-hour instructor digital Solo Endorsement verification).

2.C — Missing Portals, Multi-Tenant Routes & UI Workspaces (04 & 11 .md)

Missing Route / Screen	Target Path in .md Spec	Detailed UI & Functional Specification from 04 & 11 .md
Global Multi-Tenant Pilot Hub	/my/flights, /my/warnings, /my/certificates, /my/logbook, /my/organizations	Cross-academy SSO portal allowing freelance instructors, examiners, and students enrolled at multiple ATOs to view combined schedules, upload personal certificates, and switch active tenant.
Horizontal Multi-Resource Gantt Scheduler	/a/:tenant/bookings	Sticky left resource tree (Aircraft, Simulators, Instructors, Classrooms) + horizontal 15-min increment timeline grid (Day/Week/Month zoom), drag-to-create/resize, instructor availability shading (userAvailabilityMatrix), and conflict badges.
Live Airfield Departure / Arrival Wallboard	/a/:tenant/activities/dep-arr	Dark-mode FBO dispatch television display showing Active Runway in Use (setRunwayInUse), live surface wind/METAR, chronological Departures & Arrivals, real-time elapsed airborne timers, stand/bay assignments, and overdue alerts.
Booking Conflict Resolution Center	/a/:tenant/bookings_conflict_center	Dispatcher triage workspace listing all unresolved booking_conflicts with one-click aircraft tail swap, instructor reassignment, or Accountable Manager regulatory override (overrideBookingConflict with mandatory audit reason).
4-Tab Post-Flight & CBTA Debriefing Wizard	/a/:tenant/bookings/:id/register	Tab 1: 4-point times, Tach/Hobbs meters, Fuel/Oil & Day/Night landings. Tab 2: 1-5 CBTA segmented grading matrix for master_exercises & 8 Core Competencies + OBI checkboxes. Tab 3: Debrief notes & inline Aircraft Squawk toggle. Tab 4: Dual PIN/WebAuthn sign-off. Backed by crash-proof SessionStorage (persist:userCustomLecture).
CBTA 8-Axis Spider Radar Chart & Heatmap	/a/:tenant/cbta/matrix, /a/:tenant/users/:id/programs/:pid	8-axis SVG radar chart plotting student rolling average across KNO, FPA, FPO, COM, WLM, SAW, PRB, LST against target baseline (4.0) and safety floor (2.0), plus academy-wide maneuver bottleneck heatmaps and Stage Check queue (/a/:tenant/stage_checks).
Aircraft Squawks & CAMO CRS Work Orders	/a/:tenant/aircraft/:id/remarks, /a/:tenant/camo/work_orders	Digital aircraft technical logbook for reporting defects (info, warning, grounding_defect), deferring items to the Minimum Equipment List (MEL) with expiry dates, and recording licensed CAMO engineer Certificate of Release to Service (CRS) sign-offs.
Automated Customer Invoicing (ACI) Suite	/a/:tenant/invoicing/dashboard, /unbilled, /invoices, /customers, /price_lists	Complete financial management portal: unbilled flight activities queue, wet/dry aircraft hourly rate cards, customer prepayment retainer balances, batch PDF invoice generation, revenue analytics, and Stripe card charging.
Safety Management System (SMS) 5x5 Portal	/a/:tenant/sms/reports, /a/:tenant/sms/statistics	Confidential & standard occurrence reporting with interactive ICAO Doc 9859 5x5 Risk Matrix (Severity 1-5 x Likelihood A-E), Red/Amber/Green classification, Root Cause Analysis, and CAPA mitigation tracking.
Official EASA / FAA Electronic Logbook Export	/a/:tenant/logbooks/:user_id, /a/:tenant/logbooks/:user_id/export	Multi-column regulatory pilot logbook (SEP/MEP, Day/Night, PIC/Dual/Instructor/SPIC/PICUS, IFR approaches, FNPT Sim) with page-by-page brought-forward totals and SHA-256 digitally signed PDF export.

2.D — Missing RBAC Roles, Realtime Channels & External Integrations (06, 07, 08 .md)

- 3 Missing Operational Roles & Multi-Role Support (07_ROLES_PERMISSIONS_AND_PORTALS.md): Your app supports 4 roles (admin, instructor, ground, student) with a single role per user. The REA specification defines 7 roles—adding staff (Operations Coordinator / Dispatcher with full Gantt, conflict, certificate, and ACI billing rights), crew (Commercial Charter Pilot), and renter (Self-Hire Licensed Pilot)—and supports multiple simultaneous roles per user via UNIQUE(tenant_id, user_id, role_id).

- Dedicated Realtime WebSocket Channels & Crash-Proof Debrief Buffer (06 .md): While your app uses a single tenant-scoped Supabase Realtime subscription across tables, the REA spec defines 4 specialized real-time streams (DepartureArrivalChannel, BookingChannel, NotificationChannel, AdminAuditChannel) plus the persist:userCustomLecture SessionStorage mirror that preserves in-progress flight debriefings across iPad browser crashes.

- External Integrations (08_AGENT_SKILLS_AND_INTEGRATION_LINKS.md): Missing integrations cataloged in 08 .md: (1) Live ADS-B / SpiderTracks / RadarBox telemetry bridge that auto-transitions airborne flights to 'landed' when groundspeed drops below 30 kts inside the destination aerodrome geofence; (2) Stripe ACI payment processing; (3) S3 / Supabase Storage presigned document uploads for medical/license scans; (4) WebAuthn FIDO2 hardware 2FA.

SECTION 3 OF 3 • BLIND SPOTS ACROSS BOTH SYSTEMS

3. What Is MISSING in BOTH Your App AND the FlightLogger .md Files

Critical real-world flight academy, flight dispatch, Indian DGCA/GST, and avionics capabilities absent from both specifications

Even when combining 100% of your current SilverArrows codebase with all 13 FlightLogger REA .md files, the following 8 high-value aviation academy and flight operations capabilities are missing or under-specified in BOTH systems:

#	Capability Missing in BOTH	Operational Domain	Why It Is Required & How We Should Engineer It
3.1	**Interactive Weight & Balance (Mass & Balance) Envelope Calculator & Load Sheet**	Pre-Flight Dispatch Safety	Neither your app nor the 32-table schema in 03_DATABASE_SCHEMA_32_ENTITIES.md includes a weight_balance_stations or flight_load_sheets table or CG envelope polygon validator. Before transitioning a flight from briefing -> flight, pilots must enter crew weights, baggage weights, and fuel load to plot Zero-Fuel Weight (ZFW), Takeoff Weight (TOW), and Landing Weight (LDW) inside the aircraft's structural CG envelope (C172S, DA40, PA28, DA42), blocking dispatch if out of limits.
3.2	**Takeoff & Landing Runway Performance Calculator (Density Altitude & Crosswind)**	Pre-Flight Dispatch Safety	Your app fetches live temperature, QNH, and wind vectors, but neither system computes Pressure Altitude, Density Altitude, headwind/crosswind components against the aircraft's max demonstrated crosswind limit, or factored Takeoff Distance Required (TODR) / Landing Distance Required (LDR) vs available runway length.
3.3	**Interactive Cross-Country Operational Flight Plan (OFP) & Wind-Triangle NavLog**	Cross-Country Training	For Cross-Country (XC) syllabus lessons, neither system includes a waypoint-by-waypoint Navigation Log (NavLog) generator computing True Track, Wind Correction Angle (WCA), Magnetic Heading, Groundspeed, ETE, Fuel Burn + 45-min VFR / IFR reserve requirements, and live NOTAM decoding.
3.4	**Indian DGCA eGCA Direct Sortie Batch Exporter & FTO Monthly Return Generator**	Indian DGCA Compliance	SilverArrows operates in the Indian aviation ecosystem (VIDP, +91, DGCA Computer No, eGCA ID, SPL/FRTOL/RTR), whereas the FlightLogger REA .md files target European EASA rules. Neither system generates the standardized DGCA eGCA sortie batch upload JSON/CSV payload or FTO monthly regulatory return.
3.5	**Indian GST Aviation Tax Engine (CGST / SGST / IGST & SAC 999293 / 9964)**	Finance & Tax Compliance	The REA invoicing specification (Table 32) only models a single European VAT percentage column (tax_rate_percent). Indian aviation academies require GSTIN validation, HSN/SAC service codes, and split CGST + SGST (intra-state) vs IGST (inter-state) tax breakdowns on all student and charter invoices.
3.6	**Flight Data Monitoring (FDM / FOQA) Garmin G1000 SD Card CSV Telemetry Analyzer**	Safety & Debriefing	Modern glass-cockpit trainers (Cessna 172S G1000 NXi, Diamond DA40/DA42) log 1Hz flight parameters to SD card CSV files. Neither system includes a G1000 CSV parser to visualize the flight path during debriefing or automatically detect safety exceedances (hard landings > 2.1G, bank > 50 deg, Vne overspeed, unstabilized approaches).
3.7	**CAMO Spare Parts Inventory, Rotable Serial Traceability & Tool Calibration Store**	CAMO Maintenance Stores	While the .md files track maintenance_items and aircraft_remarks, neither system includes a spare_parts_inventory or calibrated_tools table for tracking oil filter/tire stock levels, EASA Form 1 / DGCA CA Form 1 part serial numbers, shelf-life expiry dates, or torque-wrench calibration expiry dates.
3.8	**True Offline-First Apron/Cockpit PWA with IndexedDB Multi-Flight Sync Queue**	Mobile Apron Reliability	The REA spec only caches a single active debriefing draft in window.sessionStorage (which vanishes if the tab closes). Instructors operating on remote unpaved airstrips or apron dead-zones need an IndexedDB / Service Worker offline mutation queue capable of queuing multiple briefings, block times, and debriefings offline and syncing automatically upon reconnection.

Recommended 4-Phase Implementation Roadmap

Phase	Implementation Scope	Key Deliverables
Phase 1	**Database & Schema Expansion (32 Entities + DGCA Preservation)**	Add missing tables (aircraft_models, aircraft_remarks, certificate_types, user_certificates, certificate_warnings, master_exercises, core_competencies, user_exercises, graded_competencies, landings, electronic_logbook_entries, booking_conflicts, ACI billing tables, sms_reports) while retaining all existing SilverArrows tables and DGCA fields.
Phase 2	**Core State Machines & Regulatory Dispatch Engines**	Implement the 7-Point Pre-Flight Dispatch Gating Engine (including 7/28/365-day rolling FTL & 90-day currency), the 6-Stage Flight Lifecycle FSM (scheduled -> briefing -> flight -> landed -> debriefing -> completed), and the 6-Way Atomic Post-Flight Commit (fn_register_flight) with crash-proof debrief persistence.
Phase 3	**High-Priority Operational UIs & CBTA Analytics**	Build the Horizontal Multi-Resource Gantt Scheduler, Live Airfield Departure/Arrival Wallboard, Booking Conflict Resolution Center, 4-Tab Post-Flight & CBTA 1-5 Debriefing Wizard, 8-Axis CBTA Spider Radar Chart, and Aircraft Squawk / CAMO Tech Log.
Phase 4	**Finance (ACI + Indian GST), Safety (5x5 SMS) & Dispatch Extras**	Build the Automated Customer Invoicing (ACI) suite with Indian GST (CGST/SGST/IGST), ICAO 5x5 Safety Management System (SMS) portal, Interactive Weight & Balance CG Envelope Calculator, and Signed EASA/DGCA Electronic Logbook PDF Exporter.

SECTION 4 OF 4 • POST-IMPLEMENTATION VERIFICATION & STATUS AUDIT

4. What Has Now Been Implemented & Tested vs. What Remains External

Complete verification matrix of the 32-entity REA expansion, core engines, UI routes, test suite results, and external hardware/API items

4.A — Full Implementation & Automated Test Verification Matrix (100% Implemented in App)

All core FlightLogger REA schemas, state machines, regulatory dispatch gates, CBTA 1–5 grading workflows, ACI GST invoicing, ICAO 5×5 SMS risk portals, and operational screens have been implemented in the codebase while preserving 100% of existing SilverArrows features and creating a full rollback archive (/backups/pre-flightlogger-rea-backup.tar.gz).

Subsystem / Engine	Implemented Files & Routes	Verification & Automated Test Status
1. 32-Entity Relational Schema & Seed Normalizer	src/lib/types.ts, src/lib/rea-seed.ts, src/lib/demo-storage.ts, supabase/migrations/20261030220000_flightlogger_rea_32_entities.sql	IMPLEMENTED & VERIFIED — All 32 FlightLogger REA entities typed, seeded, normalized on localStorage load, and exposed in Live Database Inspector (/database).
2. 7-Role Multi-Role RBAC Matrix	src/lib/types.ts (Role), src/lib/logic.ts (can, access, canAccess, hasRole)	IMPLEMENTED & VERIFIED — Supports admin, staff, instructor, crew, ground, student, and renter roles + secondaryRoles[] across all routes.
3. Engine 1: 6-Stage Flight Lifecycle FSM & ADS-B Bridge	src/lib/logic.ts (transitionFlightFsm, processAdsbTelemetryUpdate), /operations	IMPLEMENTED & VERIFIED — Enforces scheduled -> briefing -> flight -> landed -> debriefing -> completed, blocks airborne cancellation, and auto-lands at <30 kt.
4. Engine 2: 6-Way Atomic Post-Flight Commit (fn_register_flight)	src/lib/logic.ts (registerFlightAtomic), /cbta	IMPLEMENTED & VERIFIED — Atomically updates Fleet/CAMO + auto-grounding, CBTA 1-5 norm grading + repeat_required lock, Dual SHA-256 Electronic Logbook rows, EASA FTL 12h rest lock, ACI + Indian GST billing, and SMS auto-triggers.
5. Engine 3: 7-Point Pre-Flight Dispatch Gating Engine	src/lib/logic.ts (evaluateDispatchGates, easaRollingFtlStatus, endorseSoloFlight, overrideBookingConflict), /operations	IMPLEMENTED & VERIFIED — Evaluates Medical/SPL/FRTOL expiry, 90-day currency, Class Rating (SEP/MEP), Maint hours vs planned flight duration, 12h FTL rest, 7d/28d/900h/1000h rolling FTL ceilings, and 24h Solo Endorsement.
6. Live Ops Wallboard, Gantt Scheduler, Conflict Center & W&B	src/routes/_shell.operations.tsx (/operations)	IMPLEMENTED & VERIFIED — Dark-mode Dep/Arr Wallboard, Active Runway selector, Multi-Resource Gantt timeline, Conflict Center override modal, 24h Solo Sign-Off, and Weight & Balance CG Envelope + Runway TODR/LDR Calculator.
7. 4-Tab CBTA Debriefing Wizard, 8-Axis Radar & Signed Logbook	src/routes/_shell.cbta.tsx (/cbta), src/lib/debrief-persistence.ts	IMPLEMENTED & VERIFIED — Crash-proof SessionStorage buffer (persist:userCustomLecture), 1-5 CBTA maneuver & 8-competency grading, 8-axis SVG Spider Radar Chart, and EASA/DGCA Dual Signed Electronic Logbook PDF export.
8. Automated Customer Invoicing (ACI) & Indian GST Suite	src/routes/_shell.invoicing.tsx (/invoicing), generateInvoiceFromUnbilled()	IMPLEMENTED & VERIFIED — Unbilled flight queue, customer retainer balances, SAC 999293/996751 CGST/SGST/IGST tax invoice generator, downloadable GST Invoice PDF, and Stripe/Retainer settlement.
9. ICAO Doc 9859 Safety Management System (SMS) 5x5 Portal	src/routes/_shell.sms.tsx (/sms), classifySmsRisk()	IMPLEMENTED & VERIFIED — Interactive 5x5 Severity (1-5) x Likelihood (A-E) risk matrix heatmap, Red/Amber/Green classification, confidential Just Culture reporting, and CAPA workflow.
10. Global Pilot Hub (/my-hub) & CAMO Squawk / CRS Tech Log	src/routes/_shell.my-hub.tsx (/my-hub), src/routes/_shell.fleet.tsx (/fleet)	IMPLEMENTED & VERIFIED — Multi-tenant organization switcher, certificate expiry vault, FIDO2 WebAuthn passkey enrollment, Pilot Squawk Sheet, and CAMO Certificate of Release to Service (CRS) sign-off.

4.B — What Is Still Missing (External Production Hardware, Live Gateways & Phase-2 Extensions)

Remaining External Item	Current In-App State	What Is Needed for Live Production Deployment
1. Live Hardware ADS-B / SpiderTracks Feed	processAdsbTelemetryUpdate() engine & interactive Touchdown Simulator active in /operations	Connect a live ADS-B Exchange / FlightAware / SpiderTracks webhook secret key to stream physical aircraft GPS groundspeed packets into processAdsbTelemetryUpdate().
2. Live Stripe Merchant Secret Key & Webhook	Full ACI GST invoicing, retainer ledger & PaymentIntent simulation active in /invoicing	Configure production STRIPE_SECRET_KEY and webhook endpoint to process live credit card / UPI charges in addition to academy retainer deductions.
3. Live SendGrid / Twilio SMS & Email Gateways	In-app real-time Notification Bell, audio chime & priority feed active across all roles	Configure SENDGRID_API_KEY and TWILIO_AUTH_TOKEN to dispatch external SMS/email copies of slot alerts and certificate expiry warnings.
4. Garmin G1000 SD-Card CSV FOQA Parser & CAMO Parts Store	Cataloged in Section 3 (Missing in Both Spec & App)	Optional Phase-2 avionics telemetry CSV uploader (1Hz G1000 flight path replay) and physical spare-parts / tool calibration warehouse barcodes.